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carlin@nyc:~$ ./internship-review
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Beyond the Code

What NYC and the AI Era Taught Me

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Universal Processing · Summer 2026

\$ whoami



name	Carlin Hou
role	Software Engineering Intern
team	Universal Processing
current	New York City

// background

**My background
combines business and
information systems.**

I connect technical decisions with users,
operations, and company goals.

Taiwan → New York City: The environment changed quickly; my willingness to adapt did not.

My Best Work Starts Early

// why the early start

Morning energy goes to the work that needs the clearest thinking.

I'm usually in before 8:30—and often first at the office.

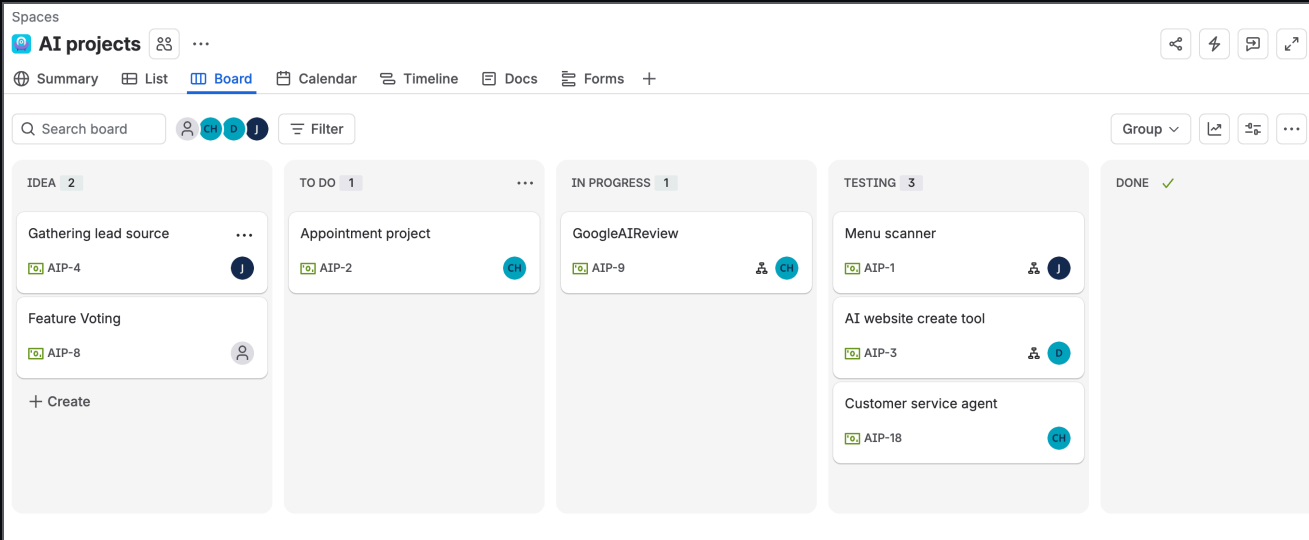
BEFORE 08:30	sync()	Taiwan handoff Email · chat · Jira · daily plan
MORNING	focus()	Deep work Demos · docs · architecture · code
LUNCH	reset()	Recharge Weekend meal prep · short nap
AFTERNOON	verify()	Close the loop QA · docs · code review · plan check

— Align early. Build deeply. Verify before leaving.

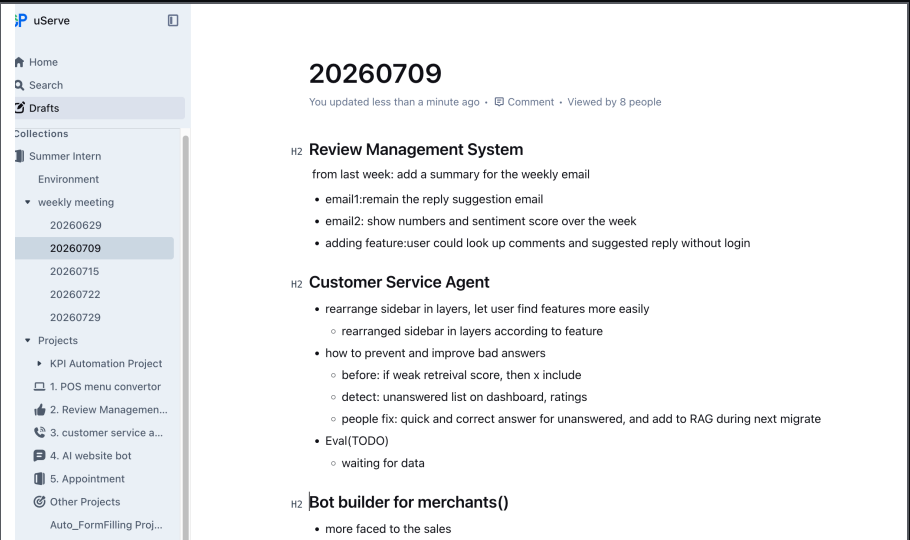
The Work Started Before the Code

// product loop

context |> requirements |> jira |> prototype |> test |> feedback |> iterate



Jira board · requirements become visible, trackable work



Weekly notes · decisions, feedback, and next steps

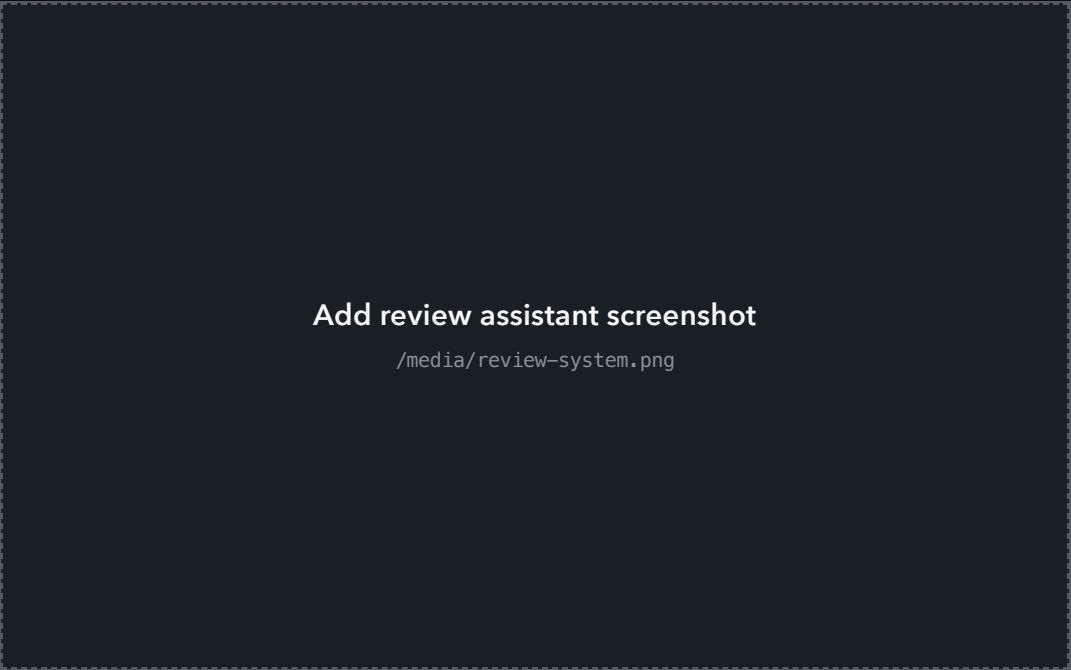
Development starts with shared understanding.

Each Project Started With a Real Bottleneck

01	<code>generate()</code>	PROBLEM	Replies take time and still require judgment.
	AI Review Assistant	SOLUTION	AI drafts; a person reviews.
02	<code>retrieve()</code>	PROBLEM	Answers are scattered across documents and past cases.
	Internal Support Chatbot	SOLUTION	Search trusted knowledge before answering.
03 / MVP	<code>coordinate()</code>	PROBLEM	A booking depends on staff, time, rooms, and equipment.
	Salon Booking Assistant	SOLUTION	Check every rule before proposing a slot.

— The common goal: less repetitive work, faster answers, and fewer manual checks.

Reduce Reply Work Without Giving Up Control



// problem → solution → value

PROBLEM	Replying across locations is repetitive, but every response still represents the business.
SOLUTION	Retrieve the review, draft a response, then keep a person in control of the final reply.
VALUE	Less first-draft effort, with judgment, tone, and accountability preserved.
retrieve() → draft() → review() → publish()	
CONTRIBUTION	OAuth · review retrieval · filters · AI/edit history
NEXT	Approval validation · multi-client workflow

Turn Scattered Knowledge Into Faster Answers

// problem → solution → value

PROBLEM Useful answers were spread across documents, project notes, and past cases.

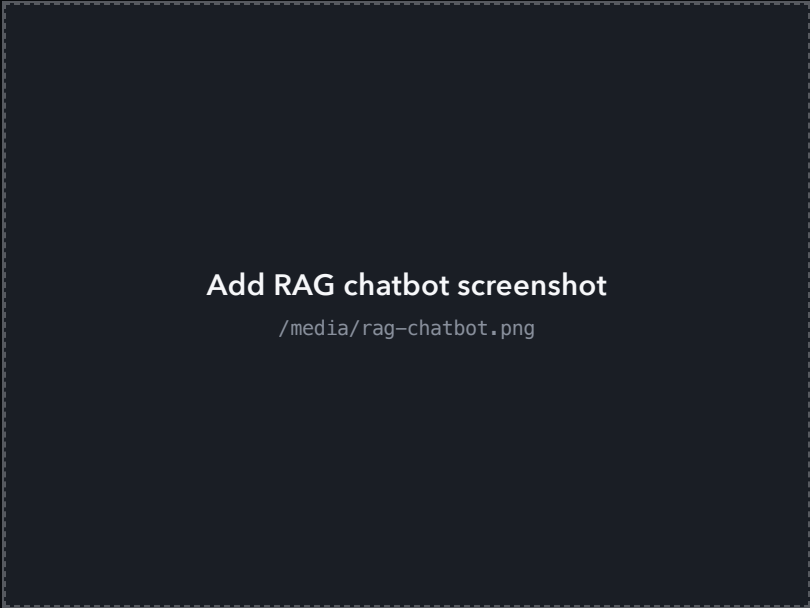
SOLUTION Search company knowledge first, then answer from the retrieved evidence.

VALUE A faster path to trusted answers, with less repeated searching.

question → retrieve(evidence) → answer()

CONTRIBUTION Knowledge integration · hybrid retrieval · evaluation

NEXT Tune retrieval with internal-user feedback



Add RAG chatbot screenshot

/media/rag-chatbot.png

Existing team knowledge becomes easier to reuse.

Prevent Booking Conflicts Before They Happen

PROBLEM	A free time slot can still fail when staff, rooms, or equipment conflict.	SOLUTION / MVP	Check the booking rules together before suggesting a time.
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service.duration

employee.skills

staff.schedule

multiService[]

add0ns

Add MVP interface screenshot · mock data

/media/salon-booking.png

room

chair

equipment

planned.calendar

planned.notifications

— MVP goal: fewer manual checks and less back-and-forth before confirming a booking.

How My Use of AI Changed

v1.0

College

AI as a search tool

Samples + Stack Overflow.

v2.0

Internship in Taiwan

AI as a coding partner

Specific tasks. Limited context.

v3.0

New York City

AI inside a complete
product process

Specs. Steps. Review. Feedback.

— The developer's role is shifting from producing more code to making better decisions.

The Bottleneck Moved.

BEFORE	IN THE AI ERA
Writing code	Defining the right problem
Implementation speed	Clear requirements
Technical execution	Human judgment
Feature delivery	Product alignment
Individual output	Shared knowledge

— Speed creates value only when the team shares a clear destination.

This summer changed how I think
about building useful products.

**Still learning.
Still building.
Still halfway.**

With thanks to Taipei City, Universal Processing, the engineers, coworkers, and
interns who shaped this summer.

Thank you.

